

Recommandations on Data Cleaning, Preprocessing, and Merging

Cleaning and Preprocessing:

- When dealing with spatial data it is recommended to fill the missing values with geographically close (neighbours) data.
- When dealing with temporal data it is recommended to fill the missing values with the data of the previous or/and next time units.
- When working on a simple (single) dataset, the feature scaling (normalization/standardization) comes at the end of the preprocessing, But, when creating a data warehouse / merging multiple datasets, the features scaling comes after the final merge, because the data distribution may change.

Fire dataset:

The target is the class fire/no fire and this feature doesn't exist in the actual dataset because it only contains fire samples.

The features in that dataset are the ones used to determine that there is a fire at a given position (long, lat), and some other features that describe the nature and confidence of the information

- Study the meaning of each feature
- Use the features (Type, Confidence) that describe the nature and confidence of the information to filter the data, or don't use them and keep all the fire...
- Eliminate useless features
- Do the EDA of the remaining features
- Create the class feature, initially all as 1 (fire)
- Generate an even distribution of points all over the study area where there is no fire as 0.
- Make sure to have an unbalanced dataset (between 70/30 and 90/10) for this PW.

Land Cover dataset:

The dataset contains polygons of small areas with one main information which is the land cover type that is codified with 2 codification systems, plus additional information.

- Study the meaning of each feature
- Choose the most precise land cover codification
- Eliminate useless features (like Area for the fire task)
- Do the EDA of the remaining features → Cleaning → preprocessing

Climate dataset:

The dataset contains 3 features, but a same position (long, lat) is repeated 12 times, one for each month. So we have Temperature_min (tmin), Temperature_max (tmax), and Precipitation (Prec) for each (long, lat) position for each month. But we are not dealing with

a time series dataset, so we need to transform it to eliminate the timing aspect from that dataset while keeping as much information as possible.

- Pivot the 12 months of each (long, lat) position into 12 columns for each feature
- Aggregate the 12 columns of each feature of each position, here you have options:
 - 1- Aggregate by calculating the Mean, Min, Max, and Std of each feature
 - 2- Aggregate by calculating the mean of each season of each feature
 - You should have 4 columns per feature, so a total of 12 columns instead of 36.

Or you can keep it as it is and use it later to perform the match with fire acquisition date...

Or you may have other interesting ideas

→ Do the EDA of the remaining features → Cleaning → preprocessing

Elevation dataset:

The dataset contains mainly one information of the elevation relative to the sea level, so it can be positive or negative.

→ Do the EDA of the remaining features → Cleaning → preprocessing

Soil dataset:

The data is represented on 2 separated files, so you have to join the soil characteristics from the database to the raster file that contains the positions (long, lat).

- Get the features from Table: HWSD2_LAYERS
- Only keep the values of LAYER = D1 (which means the characteristics of the first 20cm depth of the soil).
- Extract the features: "COARSE", "SAND", "SILT", "CLAY", "TEXTURE_USDA", "TEXTURE_SOTER", "BULK", "REF_BULK", "ORG_CARBON", "PH_WATER", "TOTAL_N", "CN_RATIO", "CEC_SOIL", "CEC_CLAY", "CEC_EFF", "TEB", "BSAT", "ALUM_SAT", "ESP", "TCARBON_EQ", "GYPSUM", "ELEC_COND".
- Join these data with the ones from the raster file according to the HWSD2_SMU_ID
- Do the EDA of the remaining features → Cleaning → preprocessing

Merge:

→ Start with the completed fire dataset and merge it with the other datasets one by one, each time using the resulting dataset to merge with the next dataset.

→ Each dataset has its own resolution, so you have to approximate the points of each dataset to the ones of the merged dataset in construction, for that you have multiple strategies:

- Take the nearest point
- Take the average of the X nearest points
- Combine when a point is within a polygon
- Convert each dataset resolution
- ...etc

What comes next:

Now you have the final merged dataset, you're gonna have to deal with:

- The imbalance
- The feature engineering (eliminate target-irrelevant features.)
- Trying different parameters for the k-NN methods
- Evaluate them all

→ Choose 2 data oriented methods to overcome the imbalance issue on k-NN and/or Decision Trees. We saw:

- Undersampling (Random undersampling, TOMEK undersampling)
- Oversampling (Random oversampling, SMOTE oversampling, ADASYN oversampling)
- Combinations

→ Compare the classical Random Forest and the balanced Random Forest